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Reading-frame constraints and retained RG content in NR4A3 fusion models of extraskeletal myxoid chondrosarcoma

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ABSTRACT

Extraskeletal myxoid chondrosarcoma (EMC) is a translocation sarcoma driven by an NR4A3 fusion, usually with the FET-family gene EWSR1. A recent report describes FET fusion oncoproteins as disrupting physiologic DNA repair, with recruitment to laser-induced double-strand breaks tracking retained FET RGG-rich content; EMC is the untested fourth transcription-factor-partner class there. The reported EMC junctions are compiled from published sources, reviews included, translated at transcript rather than coding-sequence level, and placed on that axis. Four sourced junctions (EWSR1 exons 12, 7 and 13; TAF15 exon 6) yield in-frame reading frames retaining the complete NR4A3 moiety. The EWSR1 exon 7 to NR4A3 exon 2 junction inserts 177 nucleotides in the EWSR1 frame, encoding 59 residues absent from this programme's earlier protein-level model. Retained EWSR1 RG dipeptides place type 1 at 8 of 30 and type 2 at 0 of 30, within the 0.000 to 0.267 span of the three reported EWSR1::ATF1 breakpoints: reported breakpoints, not the measured construct, whose breakpoint is unstated. That report establishes measured anchors at 0.000 and 1.000. TCF12, a minority non-FET 5' partner in EMC, falls outside the FET compositional range at every prefix on the evaluated grid (50 aa upwards in 10-aa steps), where no TCF12 prefix reaches the lowest value any FET prefix takes. No experiment was performed: sequences come from public reference transcripts on one annotation source, and each junction needs verification against a sequenced breakpoint. Five predictions with explicit falsifiers, four constructs and four controls are specified in advance.

Keywords extraskeletal myxoid chondrosarcoma; EWSR1::NR4A3; NR4A3 fusion; FET fusion; TAF15; TCF12; reading frame; RGG; double-strand break

1. Introduction

EMC is an ultra-rare sarcoma defined by rearrangement of NR4A3. The commonest 5' partner is EWSR1, a member of the FET family alongside TAF15 and FUS; a minority of cases carry TAF15, FUS or the non-FET partner TCF12. A 2025 comprehensive review states that no clinically validated agent directly targets NR4A3, and reports the systemic options as an anthracycline backbone with a low objective response rate and pazopanib at an objective response rate of 18 per cent with a median progression-free survival of 19 months (NCT02066285) [2]. The driver itself is untreated, so a candidate vulnerability that does not require the driver to be drugged is worth the cost of establishing.

A peer-reviewed report in *Cancer Research* proposes a shared lesion across FET fusion sarcomas: the chimeric protein retains the FET N-terminal low-complexity region and loses most of the C-terminal RGG-rich repeats, and that loss changes its behaviour at DNA double-strand breaks [1]. The readout is accumulation of a GFP-tagged protein at a laser-induced stripe. The report also builds an internal dose series, reintroducing one or three RGG-rich domains into EWSR1-FLI1 and into EWSR1-ATF1, and finds earlier recruitment and higher overall recruitment as the dose rises.

Three transcription-factor-partner classes were examined in that work. EMC is a fourth, and no NR4A3 fusion has been placed in the assay. The unit of work in that assay is a GFP-tagged open reading frame, so adding EMC requires new plasmids rather than a new instrument or a new analysis; what is missing is the sequence, the placement and the criteria.

A prior-art screen supports that reading of the record. A Europe PMC sweep of 322 EMC-linked records, of which 238 were retrieved as full text, was screened for ATR and replication stress and returned no hits (`emc-prior-art-2026-08-09.json`). The screen matched titles and abstracts rather than full text, so that zero establishes only that nothing is indexed on the pairing, and not that no such experiment has been done: a result inside a supplementary table of a larger FET-fusion paper would be invisible to it.

This report supplies those three things. It compiles the reported EMC junctions from published sources, primary reports and reviews alike, computes the protein each produces, places EMC on the published dose axis, classifies the one non-FET partner, and fixes five predictions with their falsifiers before any experiment.

2. Methods

2.1 Gene models and junction arithmetic

A reported fusion is an mRNA exon junction rather than a protein junction. Constructs are therefore assembled at the cDNA level, taking 5' partner cDNA from the transcript start through the end of the named exon, joining 3' partner cDNA from the start of its named exon, and translating from the 5' partner's own start codon.

The distinction changes an answer here. NR4A3 transcript exons 1 and 2 are entirely non-coding and exon 3 carries both 5' untranslated sequence and the start codon. A coding-sequence splice discards that untranslated segment; a transcript-level splice translates it in the 5' partner's frame, which is the only way to establish whether a reported junction is in frame at all.

Canonical Ensembl transcripts were used throughout. Each gene model carries four assertions: exon lengths sum to the cDNA, coding nucleotides sum to the coding sequence, the cDNA slice at the 5' untranslated boundary equals the coding sequence, and the coding sequence translates to the reference protein. Each construct carries three further self-checks: the reading frame opens with the 5' partner's N-terminus, it ends with the 3' partner's C-terminus, and both hold together. A construct failing them is reported as failing, and its sequence is withheld.

2.2 The retained-RG axis

The axis is retained RG dipeptides of the 5' FET partner as a fraction of that partner's wild-type total. It is threshold-free: an RG dipeptide either falls inside the retained segment or it does not. It is not a count of RGG domains, which depend on a box definition: the source names three RGG-rich domains in EWSR1 while the operational box-finder used here merges them into two on the same sequence. The underlying RG count requires no such definition, and box counts are reported as context only.

2.3 TCF12 comparison

Three independent tests, plus one sweep that removes a dependency on an unpinned breakpoint. Test one is the N-terminal [S,Y,G,Q] fraction over a 250-residue window, the FET prion-like signature, for TCF12 against all three FET proteins. Test two is RG dipeptide content, whole-protein and N-terminal, with the operational box count. Test three is N-terminal sequence identity by Needleman-Wunsch alignment, with the three FET-versus-FET pairs computed by the identical call as the positive control, since a single identity value in isolation carries no scale. The sweep computes the [S,Y,G,Q] fraction of every N-terminal prefix from 50 residues to full length in 10-residue steps, on that one evaluated grid, for TCF12 and for each of the three FET proteins rather than for TCF12 alone. The published version of this test swept TCF12 alone and compared its best value against the FET proteins at a single fixed 250-residue window, which is asymmetric. No sweep was run for this revision: the symmetric result, its grid, its per-protein prefixes and the resulting 0.039 gap are read from the retained artifact `emc-fet-frame-and-composition.json` under `composition.symmetric_prefix_sweep`. The conclusion therefore rests neither on one assumed junction nor on one fixed window.

2.4 Tag orientation

Tag orientation was left open. The source is internally inconsistent, its methods writing EWSR1-FLI1-GFP and its Fig. 5 legend writing GFP-EWSR1-FLI1, and a tag can itself perturb an intrinsically disordered region. The artifact therefore emits the untagged reading frame, and EMC constructs should be built in whichever orientation the recipient laboratory's existing EWSR1-FLI1 construct uses.

2.5 Reproduction

```
python3 research/modalities/emc_fet_construct_designs.py --check
python3 research/modalities/emc_fet_frame_and_composition.py --
check
python3 research/manuscripts/figures/emc_fusion_frame_figure.py --
check
```

The first command re-derives the per-construct assembly coordinates, the reading frames and the recruitment-axis rows offline from the committed input cache (`emc-construct-inputs.json`) and prints `REPRODUCES`; it does not produce the frame rule, the type-2 seam arithmetic or the composition results. The second re-derives those into `emc-fet-frame-and-composition.json`, printing `REPRODUCES` when a fresh derivation matches the committed artifact and `DRIFT`, with a non-zero exit, when it does not; its unit tests are `test_emc_fet_frame_and_composition.py`. The third compares the committed provenance stamp (`emc-atr-figure-provenance.json`) against the artifacts Figure 1 was drawn from, printing `PROVENANCE MATCHES` or `STALE`; run without `--check` it redraws the figure. These are the declared command signatures, quoted from the three modules; none of them was run to produce this revision, which recomputes no scientific result.

Retrieval, computation and drafting were carried out with substantial assistance from an AI coding agent operating on a version-controlled repository under the author's direction. The agent is not an author and cannot be one, and the author takes responsibility for the content. The agent's output did influence the substance of this report rather than its wording alone: the transcript-level reading frames, the recruitment-axis placement and the TCF12 classification are all computed results, and the pre-specified predictions follow from them. The author verified each by an independent route. Every breakpoint is quoted from a published source and checked against the cited record, every sequence figure is re-derivable by the command above, and every prose identifier is checked against a tracked fetch product by an automated linter. Those controls address the characteristic failure mode of the method, which is a fluent citation to a paper that does not exist.

2.6 Exon numbering and its provenance

Every exon rank in this report is a transcript exon rank: exon *n* is the *n*-th exon of the named transcript, counted from the transcript's first exon whether or not that exon is translated. The published sources quoted in section 3.1 state their breakpoints as exon numbers without stating which numbering they use, so the correspondence is asserted here and not taken from the sources. Two consequences follow and are stated rather than hidden. First, NR4A3 exons 1 and 2 are non-coding on ENST00000395097, so a coding-exon rank and a transcript exon rank differ by two for this gene: the label "NR4A3 exon 3" means the third exon of the transcript, which is NR4A3's first coding exon. Second, the arithmetic is carried at the nucleotide level throughout, so an exon label never enters a computation directly; it selects a boundary whose cumulative nucleotide count the artifact holds, and the four gene-model assertions and three per-construct self-checks are what a reader should audit.

The provenance of those boundaries is a single offline input cache, itself derived from UniProt and Ensembl records (section 7). **This correspondence has not been verified against a second, independent annotation source in this work.** No such verification was attempted here and none is claimed: the exon-to-nucleotide map, the transcript choices and the coding-exon offsets all rest on one retrieval into one cache. That is the same class of dependency that produced this programme's documented off-by-two error (section 6, item 2), which an independent re-derivation caught. A reader who needs the numbering to be right for their own case should re-derive it from their own annotation source rather than rely on the correspondence asserted here.

3. Results

3.1 Reported junctions

Table 1. Reported junctions, in transcript exon numbering, with the source of each.

Table 1

fusion	junction	counted frequency	sources
EWSR1::NR4A3 type 1	EWSR1 e12 to NR4A3 e3	commonest EWSR1::NR4A3 type in both series that typed EWSR1 subtypes: 10 tumours, the most frequent transcript [7]; 11 of the 15 fusion-positive cases [9]	[4,5,6]; expressed as "E-N, corresponding to EWSR1 (exons 1-12)-NR4A3 (exons 3-8)" [3]
EWSR1::NR4A3 type 2	EWSR1 e7 to NR4A3 e2	1 of 15 fusion-positive cases in Okamoto [9]; the retrieved Panagopoulos abstract [7] does not state a type-2 count	[4,6]
EWSR1::NR4A3 type 5	EWSR1 e13 to NR4A3 e3	named the second most common transcript in [7], in two cases	[5,7]
TAF15::NR4A3	TAF15 e6 to NR4A3 e3	the only reported coding junction; 3 of the 15 fusion-positive cases [9]; 4 of the 10 fusions in [10], which counts partner genes rather than EWSR1 subtypes	[4,5,7]; expressed as "T-N*, corresponding to the commonest TAF15 (exons 1-6)-NR4A3 (exons 3-8) fusion" [3]

The type 1 and type 2 junctions are defined verbatim in Nishio's review [4]: "The most common fusion transcript contains exon 12 of EWSR1 fused to exon 3 of NR4A3 (type 1), whereas exon 7 of EWSR1 is fused to exon 2 of NR4A3 in the type 2 fusion transcript" [4]. That sentence names type 1 as the most common and defines type 2 without making any frequency claim about it, so it establishes the two junctions and not a ranking of the two; the counted series in the table above, and not this definition, carry the frequencies. The junctions are corroborated independently by RT-PCR primer design, an EWSR1 exon 12 forward primer paired with an NR4A3 exon 3 reverse for type 1 and an EWSR1 exon 7 forward paired with an NR4A3 exon 2 reverse for type 2 [6], and by the counted series in which exon 12 to exon 3 was the most frequent transcript, in 10 tumours, and exon 13 to exon 3 the second most common, in two [7]. The TAF15 junction is reported as exclusive: "exon 6 of TAF15 is fused exclusively to exon 3 of NR4A3" [4], and "always" in a second source [5].

Three reported variants are not modelled, and the reasons differ. A rarer TAF15::NR4A3 isoform splices into a cryptic exon in NR4A3 intron 2, "thus encoding 25 additional amino acids prior to the NR4A3 ATG" [3]; the cryptic exon's sequence is in no source held here, so building it would mean inventing 75 nucleotides, and the same source reports the two isoforms as "essentially indistinguishable" for colony formation. FUS::NR4A3 has no exon-level breakpoint statement in the sources retrieved. TCF12::NR4A3 is reported only at genomic resolution, "the breakpoint affects the region of intron 5" [5], and TCF12 has several alternatively spliced isoforms. Absence of a construct is a statement about the sourcing rather than about the fusion.

3.2 Gene models and open reading frames

All four gene-model assertions pass on all five transcripts.

The reference gene models the junction arithmetic runs on, and the four assertions they satisfy, are given in Supplementary Table S1. The TCF12 length mismatch is not reconciled here. The two databases select different canonical isoforms, so a TCF12 residue number taken from the literature requires conversion before comparison with anything here. It does not reach the classification in section 3.5, whose decisive tests are compositional and are computed over every prefix on the evaluated grid (section 3.5).

Table 2. The four constructs. A slash marks the junction in the seam sequence. Extra junction residues are those encoded across the seam by neither partner's own reading frame.

Table 2

construct	5' retained	seam	extra residues	ORF	NR4A3 moiety
type 1, EWSR1 e12 to NR4A3 e3	EWSR1(1-431)	TAKAAVEWFD / DMPCVQAQYS	1	1058 aa	complete: AF-1, C4 zinc finger, LBD, C166
type 2, EWSR1 e7 to NR4A3 e2	EWSR1(1-264)	SQQSSSYGQQ / KPTAEEGSPA	59	949 aa	complete: AF-1, C4 zinc finger, LBD, C166
type 5, EWSR1 e13 to NR4A3 e3	EWSR1(1-472)	GRGMPPPLRG / DMPCVQAQYS	1	1099 aa	complete: AF-1, C4 zinc finger, LBD, C166
TAF15 e6 to NR4A3 e3	TAF15(1-161)	QRENYSHHTQ / DMPCVQAQYS	1	788 aa	complete: AF-1, C4 zinc finger, LBD, C166

All four are in frame. Each splits a codon across the junction, which is why the frame was computed at the nucleotide level rather than inferred from residue arithmetic.

The four junctions are four instances of one rule rather than four checked examples. A 5' partner exon joined to NR4A3 exon 2 or exon 3 is in frame if and only if the donor exon ends one nucleotide into a codon, that is if its cumulative coding nucleotide count is congruent to 1 modulo 3. Both acceptors give the same register, because NR4A3 exon 2 is 174 nucleotides, a multiple of three. The rule was evaluated over every donor and acceptor pair rather than over the reported junctions alone, and across EWSR1's seventeen coding exons the complete phase-1 donor set is exons 1, 4, 7, 9, 10, 12, 13 and 15.

3.3 A 59-residue insertion in the type-2 fusion

The named 3' exon of the type-2 junction, NR4A3 exon 2, is entirely non-coding, so the fusion mRNA carries 176 nucleotides of NR4A3 5' untranslated sequence downstream of the EWSR1 cut. EWSR1 coding sequence runs through exon 7 to nucleotide 793, which is 264 complete codons and one base over, and that base completes the first codon of the extension. The 59 residues therefore span 177 nucleotides, one donated by EWSR1 across the seam and 176 supplied by NR4A3, of which 174 come from exon 2 and 2 from exon 3. Read in the EWSR1 frame the segment contains no stop codon; the first residue is the hybrid codon AAG, encoding lysine at position 265 of the chimera, and the remaining 58 are NR4A3 sequence read in a non-native frame. The insertion lies between EWSR1(1-264) and NR4A3's own methionine, and the chimeric open reading frame is 949 aa. Figure 1C draws this seam. The type-2 fusion protein predicted by the canonical transcripts is therefore not EWSR1(1-264)::NR4A3(1-626), the protein-level model this programme itself used until this analysis. No source retrieved states what protein-level model the field uses. An N-terminal addition ahead of NR4A3's initiator is not itself novel: reference 3 describes a cryptic exon in NR4A3 intron 2 "thus encoding 25 additional amino acids prior to the NR4A3 ATG" in a rarer TAF15::NR4A3 isoform. What is specific here is the junction and the sequence of its insertion.

The weight of that statement is bounded. It is what the canonical transcripts predict for the reported exon junction, a computed consequence rather than an observed protein, and it requires checking against a sequenced junction before any reagent is ordered. A 59-residue difference nonetheless changes what a construct built from the published model contains.

3.4 Placement on the retained-RG axis

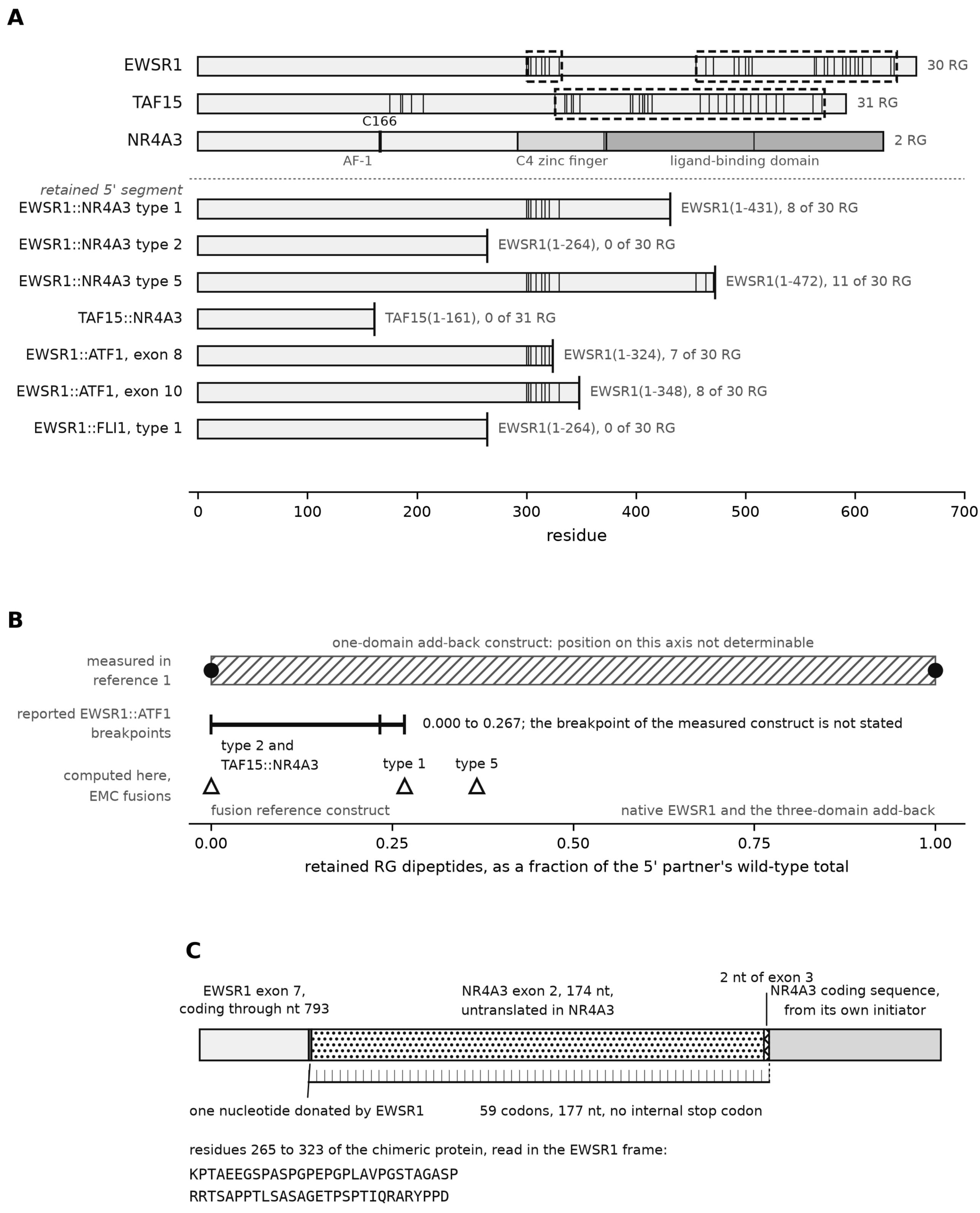

Figure 1. Reported EMC junctions on the retained-RG axis, and the type-2 seam. Three panels, drawn by `emc_fusion_frame_figure.py` and committed as `emc-fusion-frame-fig1.png` and `emc-fusion-frame-fig1.pdf`.

Panel A. Bars are drawn to residue scale on the shared axis. Each thin vertical tick marks one RG dipeptide, at the 1-based position of its arginine (`emc-fet-frame-and-composition.json`, `composition.rg_content.<protein>.rg_positions`); the count printed at the right of each wild-type bar is that protein's total. Dashed rectangles mark operationally defined RGG boxes — maximal merged runs of 60-residue windows each containing at least six RG dipeptides, trimmed to start at the first RG the box contains and to end two residues past its last RG-start (`emc_fet_idr_census.py`, `RGG_WINDOW = 60`, `RGG_MIN_RG_IN_WINDOW = 6`); the spans drawn are `rgg_boxes_operational` in `emc-fet-construct-designs.json`. This finder returns two boxes on EWSR1 and one on TAF15; the source paper names three RGG-rich domains in EWSR1, and the box definition was not tuned to reproduce that count. NR4A3 has no box because the finder returns none for it. `C166` is a printed text label placed above the NR4A3 bar with a single vertical marker drawn on that bar at residue 166; it identifies that residue position and nothing more, and no functional property is asserted for it here. Below the dotted rule, each fusion row shows the retained 5' segment ending at a heavy end-cap at its breakpoint, carrying only the RG ticks at or before that breakpoint; the right-hand text gives the retained residue range and the retained-of-total RG count. Panel A carries four weights of vertical black line — RG tick, dashed box outline, breakpoint end-cap and the `C166` marker — and the figure prints no key of its own, so the reader is directed to this caption for all four.

Panel B. The retained-RG fraction axis, carrying the fusion positions of Table 3, the hatched band labelled "position on this axis not determinable" for the construct whose reintroduced RGG-rich domain the source does not identify, and the 0.000 to 0.267 interval spanned by the three reported EWSR1::ATF1 breakpoints. That interval describes reported breakpoints; it is not a measured recruitment range.

Panel C. The type-2 seam, drawn as one nucleotide donated by EWSR1 followed by 176 nt of NR4A3 untranslated sequence: 174 nt of NR4A3 exon 2, the block labelled untranslated in NR4A3, and the first 2 nt of NR4A3 exon 3, annotated separately in the panel. Those 177 nt are the extension of 59 codons carrying no internal stop codon. The earlier caption counted only the exon 2 block and the donated nucleotide, which does not reach 177. The panel does not print the words "type 2": it draws the EWSR1 exon 7 to NR4A3 exon 2 junction, which is the only reported EMC junction using the exon 2 acceptor, and it should be read as that junction and not as a general statement about the acceptor.

What the figure shows, and what it does not. The three panels carry the reported junctions, the RG content each retains, the retained-RG fraction axis and the type-2 seam. The frame rule as a rule over all donor and acceptor pairs, the complete phase-1 donor set, the symmetric prefix sweep and its 0.039 margin, and the TAF15 zero-RG margin are results of sections 3.2, 3.4 and 3.5 and appear in no panel. Two drawing-level observations, on the legibility of the closely spaced NR4A3 ticks and on the binding between the Panel A box literals and their source data field, are recorded in the integration QA note cited in the editorial comment above rather than in this caption.

Table 3. EMC fusions and their comparators on one axis, with the status column distinguishing a position measured in reference 1 from a reported breakpoint of a disease in which the mechanism was measured. Fractions are of the 5' partner's wild-type RG total.

Table 3

construct	5' partner retained	RG kept	fraction	status
EWSR1-FLI1 (0 RGG domains), reference 1's reference fusion	EWSR1(1-264)	0 / 30	0.000	measured in reference 1
EWSR1-RGG(1)-FLI1	not identified	not placeable	not placeable	measured in reference 1; reference 1 does not identify which RGG-rich domain was reintroduced, so its retained RG count is unknown
EWSR1-RGG(3)-FLI1 and native EWSR1	full length	30 / 30	1.000	measured in reference 1
EWSR1::FLI1 (Ewing, type 1), EWSR1 e7	EWSR1(1-264)	0 / 30	0.000	a reported breakpoint of a disease in which the mechanism was measured
EWSR1::ATF1 (clear cell, reported type), EWSR1 e7	EWSR1(1-264)	0 / 30	0.000	a reported breakpoint of a disease in which the mechanism was measured
EWSR1::ATF1 (clear cell, commonest type), EWSR1 e8	EWSR1(1-324)	7 / 30	0.233	a reported breakpoint of a disease in which the mechanism was measured
EWSR1::ATF1, EWSR1 e10	EWSR1(1-348)	8 / 30	0.267	a reported breakpoint of a disease in which the mechanism was measured
EWSR1::NR4A3 type 2	EWSR1(1-264)	0 / 30	0.000	computed here; predicted
TAF15::NR4A3	TAF15(1-161)	0 / 31	0.000	computed here; predicted
EWSR1::NR4A3 type 1	EWSR1(1-431)	8 / 30	0.267	computed here; predicted
EWSR1::NR4A3 type 5	EWSR1(1-472)	11 / 30	0.367	computed here; predicted

Reference 1 built one EWSR1-ATF1 construct and the retrieved text does not state its EWSR1 breakpoint. Three EWSR1::ATF1 breakpoints are reported, retaining 0, 7 and 8 of 30 RG dipeptides, so the EWSR1::ATF1 comparator is a span from 0.000 to 0.267 rather than a point. The firmly measured positions on this axis are 0.000 and 1.000.

Figure 1A shows the retained segments and their RG content, and Figure 1B the axis itself. Type 2 sits where EWSR1-FLI1 sits, at zero, on a retained EWSR1 segment identical in sequence over the shared prefix. Type 1 sits at 0.267, inside the reported EWSR1::ATF1 span of 0.000 to 0.267. Neither EMC type falls outside the range of retained-RG fractions the reported breakpoints of the measured diseases already cover. Placing a fusion on the retained-RG axis does not establish that recruitment was measured at that position: reference 1 measured recruitment at 0.000 and at 1.000, and a position between them is a computed placement on the axis and not a measured recruitment result.

Two readings are excluded. Retaining some RG content does not predict absence of the phenotype: the commonest clear-cell type retains seven RG dipeptides and the mechanism was measured in that disease regardless, so the axis is a comparison and not a threshold. The placement of TAF15::NR4A3 was also open until this computation. TAF15's sourced exon 6 junction retains 161 residues and TAF15's first complete RG dipeptide occupies residues 175 and 176, so the junction lies inside the strict zero-RG window with 14 residues of headroom, where the earlier sweep could report only a range of 100 to 170. The largest prefix of TAF15 carrying no complete RG dipeptide is 175 residues: a prefix of 175 contains none and a prefix of 176 contains one, so the headroom is 175 minus 161.

A stored field disagrees, and the disagreement is a label rather than a measurement. The artifact carries 174 and 13. 174 is a conservative boundary: it ends one residue before the arginine that starts the first RG, so a prefix of that length cannot contain even the beginning of one. It is not the largest zero-complete-RG prefix, and 13 is not the complete-dipeptide headroom, although the field is labelled as though both were. The stored values and their history are left exactly as they are, and this paragraph is the current reading; a reader can reproduce the discrepancy by counting RG dipeptides in prefixes of 174, 175 and 176 residues of the cached TAF15 sequence.

3.5 TCF12 outside the FET compositional range

TCF12::NR4A3 was reported in one of 26 cases in one EMC series [6]. TCF12 is not a FET-family gene. The class argument therefore predicts that these cases do not carry the lesion, which is a prediction testable inside one disease on one slide.

Table 4. TCF12 against the three FET proteins.

test	the three FET proteins	TCF12	separation
N-terminal 250-aa [S,Y,G,Q] fraction	EWSR1 0.540, TAF15 0.620, FUS 0.804	0.368	decisive
symmetric [S,Y,G,Q] sweep, every prefix from 50 aa upwards in 10-aa steps, all four proteins on the same grid	lowest FET prefix value 0.439 (EWSR1, residues 1-560)	best TCF12 prefix 0.400, at residues 1-160	separates, by 0.039; no TCF12 prefix on the evaluated grid reaches the lowest value any FET prefix takes on that grid
RG dipeptides, whole protein	30, 31, 24	7	clear
RGG boxes, operational definition	2, 1, 2	0	clear
N-terminal identity, Needleman-Wunsch	FET versus FET 26.1 to 35.7 per cent	TCF12 versus FET 16.8 to 20.5 per cent	separates, modestly

TCF12 is classified as non-FET on the compositional tests. The identity result carries the qualification: 20.5 per cent against a FET-versus-FET floor of 26.1 per cent is a real gap but a modest one, which follows from the FET N-termini being low-complexity and only 26 to 36 per cent identical to each other. TCF12 has no RGG box, roughly a quarter of the RG content, and no N-terminal prefix on the evaluated grid reaching the lowest value any FET prefix takes on that grid, which makes the classification robust to the unpinned TCF12 breakpoint across the lengths evaluated. The grid is every prefix from 50 aa upwards in 10-aa steps, 66 prefixes for TCF12 and 61, 55 and 48 for EWSR1, TAF15 and FUS; lengths between grid points were not evaluated and no claim is made about them. The margin on that sweep is 0.039 and the published version of the comparison, which gave TCF12 every prefix and the FET proteins one fixed 250-residue window, overstated it.

4. Pre-specified predictions

Table 5. Predictions fixed before any experiment, each with the observation that falsifies it. P1 to P4 are held in `emc-fet-construct-designs.json` under `rgg_dose_calibration_and_predictions.registered_predictions`; P5 is held in the same file under `tcf12_negative_control.registered_prediction`.

id	prediction	falsified by
P1	EWSR1::NR4A3 type 2 is recruited to laser-induced breaks with kinetics indistinguishable from EWSR1-FLI1. Basis: 0 of 30 RG retained, on a segment byte-identical to the reference construct's	no accumulation at the stripe, or kinetics matching native EWSR1 rather than the fusion reference
P2	EWSR1::NR4A3 type 1, the commonest EMC fusion, is recruited earlier than type 2 and closest to the commonest clear-cell EWSR1::ATF1 type. Basis: 8 of 30 against 7 of 30	type 1 recruiting no earlier than type 2, which would place the variable elsewhere; or type 1 not recruited at all
P3	TAF15::NR4A3 is recruited, at the zero end of the axis. Basis: TAF15(1-161) retains 0 of 31, with 14 residues of margin to TAF15's first RG at 175	kinetics indistinguishable from native TAF15
P4	The type-1 and type-2 pair reproduces the RGG dose-dependence with no add-back construct, being two naturally occurring points on one axis in one disease with one 3' partner	the pair showing no kinetic difference, which would bound the dose-dependence to engineered constructs
P5	TCF12::NR4A3 is not recruited, behaving like the source's full-length FLI1 control, which "showed no accumulation at laser-induced DSBs" [1]	recruitment of TCF12::NR4A3

The registered text above is reproduced unchanged, including two figures this report corrects. P1 to P5 are a preregistration. Their wording, their bases and their falsifiers are reproduced here exactly as they were registered, and nothing below rewrites them; the corrections and the limitations are stated separately, as corrections, so that a reader can see both the registered claim and what is now known about it.

The margin figure in P3. P3's registered basis states 14 residues of margin to TAF15's first RG at residue 175, and that is the correct complete-dipeptide headroom: the largest prefix with no complete RG is 175 residues, the retained length is 161, and 175 minus 161 is 14. This was previously described here as one of two conventions differing by one, the other giving 13. That description was wrong. 13 is not a second convention; it is the artifact's conservative 174-residue boundary, which stops before the arginine that opens the first RG and is labelled as though it were the largest RG-free prefix. Section 3.4 records the discrepancy and how to reproduce it. P3's registered wording is unchanged, as is every stored artifact byte.

Redundancy and equivalence within the registered set. Two limitations of the set as registered are identified here and neither is repaired by editing it. P4's falsifier, "the pair showing no kinetic difference", overlaps P2's first falsifier, "type 1 recruiting no earlier than type 2", without being identical to it: the first is a two-sided absence of any difference, while the second is directional and is also satisfied by type 1 recruiting later. One experiment can therefore satisfy both at once, so the set counts fewer independent tests than its five ids suggest, but the two entries are not interchangeable. P1 as registered predicts kinetics *indistinguishable* from a comparator. Failure to detect a difference does not establish equivalence. The registered wording supplies no equivalence margin or precision criterion, so a nonsignificant comparison alone cannot confirm P1. Whether to re-register a narrowed set is a scientific decision about a preregistration and is not taken here.

Two records, and what each holds. The machine-readable record `rgg_dose_calibration_and_predictions.registered_predictions` holds four entries, P1 to P4. This manuscript carries five hypotheses, P1 to P5, with P5 held separately under `tcf12_negative_control.registered_prediction`. Both records are preserved as they stand. They are not the same record, their entries are not word-for-word identical, and neither is edited here to agree with the other; a reader auditing a prediction should read the entry in the record they cite.

P5 tests the specified non-recruitment prediction. What these patterns can carry is narrower than an earlier version of this section claimed. The predictions specify an ordering and a non-recruitment, and an ordering alone cannot assign a unique cause. Type 1 and type 2 differ in retained residues and junction segments other than RG content, so a kinetic difference between them is not attributable to RG dose by this design; and a partner-alone control shows what the partner does on its own, not that the other moiety is necessary. Four outcomes are distinguishable, and each is stated as what it is consistent with rather than what it demonstrates.

TCF12::NR4A3 not recruited while EWSR1::NR4A3 is recruited leaves P5 standing. It is consistent with a FET-specific requirement and does not establish one, because the two chimeras differ in more than the presence of a FET N-terminus. Both recruited contradicts P5's exclusive prediction; it does not show that FET-mediated recruitment is absent in the EWSR1 chimera, since a shared phenotype is not evidence of a shared cause, and it does not by itself locate the driver in the NR4A3 moiety. GFP-NR4A3 alone is run in the same experiment because NR4A3's own behaviour is a fact worth having, not because recruitment of NR4A3 alone would remove the FET attribution. TCF12::NR4A3 recruited while EWSR1::NR4A3 is not is inconsistent with the ordering the structural argument predicts. If neither is recruited, the tested constructs do not show recruitment under these assay conditions.

P5 is untested until a TCF12::NR4A3 construct exists. No such construct is emitted, for the reason in section 3.1, and full-length GFP-TCF12 does not substitute for one: it is a partner-alone control and it cannot test a prediction about a fusion.

Four things are explicitly not predicted. Retained RGG content is one input to recruitment kinetics rather than the only one; reference 1's own data show a second variable, EWSR1::ATF1 recruiting like EWSR1-FLI1 but with "differences in departure timing", and recruitment depending "at least in part" on native EWSR1, which these constructs do not control. No effect size is predicted: the axis is ordinal, earlier or later and more or less, because reference 1 reports it that way, and no slope is available to quote. Nothing downstream is predicted; the predictions concern recruitment kinetics only and say nothing about ATM signalling, ATR dependency, drug sensitivity, efficacy, safety, dosing or any clinical question. And the 3' partner is a nuclear receptor with its own DNA-binding domain: Gracilla and colleagues showed that a DNA-binding-domain mutation did not change EWSR1-FLI1's localisation, but that was measured on an ETS domain rather than a C4 zinc finger, which is why GFP-NR4A3 alone is included as a control.

5. Constructs and controls

The assay's unit of work is a GFP-tagged open reading frame. From its methods: "U2OS cells expressing EWSR1-GFP, EWSR1-FLI1-GFP, EWSR1-ATF1-GFP, EWSR1-WT1-GFP or the various mutant forms of the fusion oncoproteins were seeded in 8-well Lab Tek II Chamber Slides ... Cells were treated with 1 microgram/ml Hoechst 33342 ... for 30 minutes prior to micro-irradiation ... 5-pixel wide stripes were drawn in every cell nucleus ... and irradiated with a 405nm diode laser (40mW). Images were acquired pre-irradiation and at 1-minute intervals post-laser damage for 15 minutes" [1]. Adding EMC to that panel requires plasmids and nothing else.

Four wild-type controls anchor both ends of the recruitment axis for EMC's own partner genes, without which a delayed curve cannot be told from a poorly expressed construct.

The four controls, their roles and their predictions are given in Supplementary Table S2. No TCF12::NR4A3 construct is emitted, for the reason in section 3.1. A laboratory holding a TCF12::NR4A3 case should sequence the junction. Without one, P5 cannot be run. Full-length GFP-TCF12 answers a different and narrower question, whether full-length TCF12 reaches the damage stripe at all, and it is a partner-alone control rather than a stand-in for the fusion. A result from GFP-TCF12 leaves P5 untested.

6. Limitations

1. **These are computed designs rather than validated reagents.** Nothing here has been synthesised, expressed or sequenced. Every junction requires verification against a sequenced breakpoint before any order is placed.
2. **This analysis inherits a documented failure mode of its own method.** An earlier committed artifact in this programme, built from a stated Ensembl methodology, indexed a coding-exon offset table with transcript exon numbers; the label "NR4A3 exon 3" resolved to transcript exon 5, and all seven junctions it emitted silently deleted NR4A3's AF-1 domain and the first zinc finger of its C4 DNA-binding domain. That error survived review and was caught by re-derivation. Every boundary above therefore carries its provenance and every construct carries self-checks.
3. **Canonical Ensembl transcripts only, from a single annotation source.** A tumour may use a different transcript or a different breakpoint, in which case the exon-to-residue map changes and so does the protein. The exon numbering, the transcript choices and the coding-exon offsets all rest on one offline input cache and have not been verified against a second, independent annotation source in this work (section 2.6).
4. **The predictions concern recruitment kinetics and nothing else.** They make no claim about ATM signalling, ATR dependency or drug sensitivity, and no claim about efficacy, safety, tolerability, dosing, patient selection or clinical readiness.
5. **Retained RGG content is one input among several**, and the constructs do not control native EWSR1, which reference 1 shows contributes.
6. **FUS::NR4A3 and TCF12::NR4A3 have no sourced transcript-level junction here**, which bounds the sourcing rather than the fusions.
7. **The class inheritance is the whole of the transfer argument.** The replication-stress vulnerability was measured on other FET fusions in other diseases; no NR4A3 fusion has been tested for it, and no computation in this report changes that.
8. **No laboratory work is proposed by the author.** This programme has no laboratory. The deliverable is the design, the prediction and the criteria.

Scope of the claims. This is a sequence-analysis report. It asserts no efficacy, potency, dose, safety, therapeutic window or clinical readiness for any agent in any disease, and makes no treatment recommendation. The replication-stress vulnerability that motivates the assay is inherited from the FET fusion class and has never been measured on an NR4A3 fusion; that inheritance is the limit of what is claimed here, and no result below is EMC-specific. Every construct is a computed design for verification against a sequenced breakpoint before any reagent is ordered.

7. Data and code availability

item	location
Producer	`emc_fet_construct_designs.py`
Computed artifact holding the per-construct assembly coordinates, reading frames and axis rows	`emc-fet-construct-designs.json`
Producer of the frame rule, the type-2 seam arithmetic and the composition results	`emc_fet_frame_and_composition.py`
Its computed artifact	`emc-fet-frame-and-composition.json`
Its unit tests	`test_emc_fet_frame_and_composition.py`
Figure 1 generator and the provenance stamp it writes	`emc_fusion_frame_figure.py`, `emc-atr-figure-provenance.json`
Offline input cache the artifact re-derives from	`emc-construct-inputs.json`
NR4A3 exon audit	`nr4a3-exon-audit.json`
RG and RGG-box definitions, and the breakpoint sweep	`emc_fet_idr_census.py`, `emc-fet-idr-census.json`
Companion assessment of the class-inheritance argument	`emc-atr-vulnerability-assessment.md`
Prior-art screen, with its retrieval record and its stated limits	`emc-prior-art-2026-08-09.json`
Separate pre-registered protocol for the drug-response half of the same question, which this report does not address	`emc-atri-prereg.md`

`emc-fet-construct-designs.json` was produced by GitHub Actions run 30857647907 on `depmap-dependency.yml` in the public repository, and its producer's `--check` re-derives it offline. The frame-and-composition artifact and the figure carry their own producers and checks, listed in section 2.5. Any figure above that disagrees with the artifact it is drawn from is an error in this document.

The analysis runs on a standard processor and requires no specialised hardware, licensed software or paid service, so it can be reproduced at no cost.

8. References

1. Gracilla DE, Menon S, Breese MR, Lin YP, Dela Cruz FS, Feinberg TY, et al. FET Fusion Oncoproteins Disrupt Physiologic DNA Repair and Create a Targetable Opportunity for ATR Inhibitor Therapy. *Cancer Research* 2026;86:2660-2677. PMID 41811428. PMC13223543. doi 10.1158/0008-5472.can-25-2166.
2. Remiszewski P, Falkowski S, Szumera-Ciećkiewicz A, Spałek MJ, Rutkowski P, Czarnecka AM. From pathogenesis to the patient's bedside: a comprehensive review of extraskeletal myxoid chondrosarcoma. *J Cancer Res Clin Oncol* 2025;151(11):283. PMID 41055792. PMC12504171. doi 10.1007/s00432-025-06316-5.
3. Brenca M, Stacchiotti S, Fassetta K, Sbaraglia M, Janjusevic M, Racanelli D, et al. NR4A3 fusion proteins trigger an axon guidance switch that marks the difference between EWSR1 and TAF15 translocated extraskeletal myxoid chondrosarcomas. *J Pathol* 2019;249(1):90-101. PMID 31020999. PMC6766969. doi 10.1002/path.5284.
4. Nishio J, Iwasaki H, Nabeshima K, Naito M. Cytogenetics and molecular genetics of myxoid soft-tissue sarcomas. *Genet Res Int* 2011;2011:497148. PMID 22567356. PMC3335514. doi 10.4061/2011/497148. Source of the verbatim type 1 and type 2 exon-level definitions and of the TAF15 exclusivity statement quoted in section 3.1.
5. Cerrone M, Cantile M, Collina F, Marra L, Liguori G, Franco R, et al. Molecular strategies for detecting chromosomal translocations in soft tissue tumors (review). *Int J Mol Med* 2014;33(6):1379-1391. PMID 24714847. PMC4055444. doi 10.3892/ijmm.2014.1726. Source of the type 5 definition, the second TAF15 exclusivity statement, and the TCF12 genomic-only intron 5 breakpoint quoted in section 3.1.
6. Agaram NP, Zhang L, Sung YS, Singer S, Antonescu CR. Extraskeletal myxoid chondrosarcoma with non-EWSR1-NR4A3 variant fusions correlate with rhabdoid phenotype and high-grade morphology. *Hum Pathol* 2014;45(5):1084-1091. PMID 24746215. PMC4015728. doi 10.1016/j.humpath.2014.01.007.
7. Panagopoulos I, Mertens F, Isaksson M, Domanski HA, Brosjö O, Heim S, et al. Molecular genetic characterization of the EWS/CHN and RBP56/CHN fusion genes in extraskeletal myxoid chondrosarcoma. *Genes Chromosomes Cancer* 2002;35(4):340-352. PMID 12378528. doi 10.1002/gcc.10127. Counted series. The quotation retained in the frame-and-composition artifact gives counts without a denominator: type 1 in 10 tumours, the most frequent transcript, and type 5 in two cases, the second most common. The denominator of 15 EWS/NR4A3 cases is recorded in this repository's editorial note on this reference, not in that quotation.
8. UniProt and Ensembl reference records for EWSR1 (ENST00000397938), NR4A3 (ENST00000395097), TAF15, FUS and TCF12 (UniProt Q99081), as retrieved into the input cache in section 7.
9. Okamoto S, Hisaoka M, Ishida T, Imamura T, Kanda H, Shimajiri S, Hashimoto H. Extraskeletal myxoid chondrosarcoma: a clinicopathologic, immunohistochemical, and molecular analysis of 18 cases. *Hum Pathol* 2001;32(10):1116-1124. PMID 11679947. doi 10.1053/hupa.2001.28226. Counted series: fusion transcripts detected in 15 of 18 cases, EWS-CHN type 1 in 11, type 2 in 1 and TAF2N-CHN in 3.
10. Sjögren H, Meis-Kindblom JM, Orndal C, Bergh P, Ptaszynski K, Aman P, Kindblom LG, Stenman G. Studies on the molecular pathogenesis of extraskeletal myxoid chondrosarcoma - cytogenetic, molecular genetic, and cDNA microarray analyses. *Am J Pathol* 2003;162(3):781-792. PMID 12598313. PMC1868116. Counted series, by partner gene rather than by EWSR1 subtype: EWS-TEC five cases, TAF2N-TEC four, TCF12-TEC one, of ten tumours.

9. Supplementary tables

Three supplementary tables are proposed. S1 and S2 carry content moved from the main text without change. S3 is new to this revision and is assembled from one committed artifact; it introduces no construct that the artifact does not already hold.

Supplementary Table S1. Reference gene models, from the UniProt and Ensembl records retrieved into the input cache [8]. Transcript accessions are the ones the cache holds; the cache records its retrieval as 2026-08-12. No assembly, release or version suffix is given, because none is recorded.

gene	transcript	protein	transcript / coding exons	Ensembl matches UniProt
EWSR1	ENST00000397938	656 aa	17 / 17	yes
TAF15	ENST00000605844	592 aa	16 / 16	yes
FUS	ENST00000254108	526 aa	15 / 15	yes
NR4A3	ENST00000395097	626 aa	8 / 6, exons 1-2 non-coding	yes
TCF12	ENST00000333725	706 aa	21 / 19	no; UniProt Q99081 is 682 aa

Supplementary Table S2. Wild-type controls and their predictions. Full-length sequences are in the artifact under `wild_type_controls`.

control	role	prediction
GFP-EWSR1, full length	fast-recruitment anchor, already held by any laboratory running the assay	rapid recruitment, as published. Failure to reproduce it makes nothing else in the run interpretable
GFP-TAF15, full length	wild-type anchor for the TAF15::NR4A3 arm	rapid recruitment, as TAF15 carries its own C-terminal RGG region. Not previously reported in this assay, so a prediction rather than a reproduction
GFP-NR4A3, full length	partner-alone control, the EMC analogue of reference 1's GFP-FLI1 control	no accumulation. Recruitment of NR4A3 alone would show that the partner reaches breaks on its own; it would not by itself remove a FET contribution in the fusion, which a partner-alone control cannot isolate
GFP-TCF12, full length	non-FET partner-alone control	no accumulation, TCF12 being non-FET by section 3.5. It is not a TCF12::NR4A3 construct and does not test P5, which stays untested without one

Supplementary Table S3. Per-junction assembly coordinates for each reported junction with a sourced transcript-level breakpoint. Every cell is a verbatim copy of a recorded leaf of `emc-fet-construct-designs.json` (sha256 `726aae02ae38b41c34d4398363e3581cfc9e4602d0fe9d65906a4fba79c2048b`), a fixed column label, or `UNRESOLVED`. No value in this table was derived, inferred, rounded or interpolated, and no new construct is proposed by it.

Field	EWSR1_NR4A3_type1	EWSR1_NR4A3_type2	EWSR1_NR4A3_type5	TAF15_NR4A3
Junction (label)	EWSR1::NR4A3 type 1 — EWSR1 exon 12 :: NR4A3 exon 3	EWSR1::NR4A3 type 2 — EWSR1 exon 7 :: NR4A3 exon 2	EWSR1::NR4A3 type 5 — EWSR1 exon 13 :: NR4A3 exon 3	TAF15::NR4A3 — TAF15 exon 6 :: NR4A3 exon 3
Reported rank — LEGACY SOURCE DESCRIPTION, not a counted frequency (see note)	the commonest reported EWSR1::NR4A3 transcript type	the second reported EWSR1::NR4A3 transcript type	a minority reported type	the only reported TAF15::NR4A3 coding junction
5' transcript	ENST00000397938	ENST00000397938	ENST00000397938	ENST00000605844
5' last exon retained (transcript rank)	12	7	13	6
3' transcript	ENST00000395097	ENST00000395097	ENST00000395097	ENST00000395097
3' first exon retained (transcript rank)	3	2	3	3
CUT — 5' cDNA nt retained	1363	862	1486	570
CUT — 5' coding nt retained	1294	793	1417	484
5' UTR length (nt)	69	69	69	86
CUT — 3' cDNA resume offset (0-based)	697	523	697	697
3' UTR nt read through	2	176	2	2
Assembled cDNA length (nt)	6270	5943	6393	5477
ORF length (nt)	3174	2847	3297	2364
ORF length (aa)	1058	949	1099	788
5' residues fully encoded	431	264	472	161
Codon split across junction	yes	yes	yes	yes
Seam residue index (1-based)	432	265	473	162
Junction context (aa)	TAKAAVEWFD|DMPCVQAQYS	SQQSSSYGQQ|KPTAEEGSPA	GRGMPPPLRG|DMPCVQAQYS	QRENYSHHTQ|DMPCVQAQYS
Junction context (nt)	AATGGTTTGATG|ATATGCCCTGCG	ACGGGCAGCAGA|AGCCCACTGCGG	CACTCCGTGGAG|ATATGCCCTGCG	ACCACACACAAG|ATATGCCCTGCG
Extra junction-encoded residues	1	59	1	1
In frame (self-check)	yes	yes	yes	yes
5' start matches partner (self-check)	yes	yes	yes	yes
3' C-terminus intact (self-check)	yes	yes	yes	yes
CUT — genomic breakpoint coordinates	UNRESOLVED	UNRESOLVED	UNRESOLVED	UNRESOLVED

Supplementary Table S3, continued. Reported fusions with no sourced transcript-level junction.

Fusion	Every assembly field	Recorded status
FUS::NR4A3	UNRESOLVED	NO transcript-level junction sourced in this repo's literature cache
TCF12::NR4A3	UNRESOLVED	GENOMIC breakpoint only — reported as TCF12 intron 5, not as an mRNA exon junction, and TCF12 has several alternatively-spliced isoforms

Note on the "Reported rank" row. Its four cells are verbatim leaves of the artifact and are retained unchanged, with their provenance, but they are a legacy source description of how each junction was characterised, not a frequency counted in this work. Table 1 carries the corrected frequency evidence, with each count attached to the series it comes from; where the two differ, Table 1 governs. No artifact value was modified to add this label.

Genomic breakpoint coordinates are `UNRESOLVED` for all four junctions because the input carries transcript and cDNA coordinates only; they are not back-derived from exon ranks. FUS::NR4A3 and TCF12::NR4A3 appear in no column because the input records no transcript-level junction for either — TCF12 is reported at genomic resolution only, in intron 5, and TCF12 has several alternatively spliced isoforms. That is a limit of the sourcing and not evidence about the fusions.